

Adi Vaani: National Mission for Digital Empowerment of India's Tribal Languages

India has one of the richest linguistic landscapes in the world, with more than 443 tribal and Indigenous communities speaking hundreds of distinct languages and oral traditions. These languages contain deep cultural, ecological, and social knowledge that has been passed down over centuries. Yet many of these languages remain invisible in digital systems. This digital exclusion threatens their survival and increases the social and economic marginalization of millions of citizens. It also limits their access to e-governance services, digital education, and opportunities in the modern economy.

According to UNESCO, nearly 200 Indian tribal languages are endangered, and many could disappear by 2050 unless continued effort is made to preserve them digitally and include them in education. Most of these languages have limited written resources, lack standardized orthographies, and are almost absent from modern AI systems or digital platforms.

Mainstream languages, such as Hindi, Tamil, and Bengali, have benefited from decades of digitization and technological development. In contrast, tribal languages like Gondi, Bhili, Mundari, and Korku lack digital presence, educational resources, and computational tools. The Adivaani Mission is a strategic initiative designed to address this critical national challenge. It aims to leverage state-of-the-art Artificial Intelligence to safeguard our tribal linguistic heritage and ensure the digital inclusion of these communities. By positioning language technology as a tool for social equity and cultural sovereignty, this mission directly supports the vision of an inclusive and empowered Digital India.

1. Adi Vaani Objectives: Three Pillars of Transformation

Our objectives are organised around four interconnected strategic pillars:

1. Systematic Language Documentation and Digitization

- Expand comprehensive field documentation to a minimum of ten critically endangered low-resource tribal languages.
- Enrich the National Tribal Language Archive with robust, high-quality spoken and written corpora, creating a permanent digital record.
- Facilitate community-led development and standardization of orthographies for languages that currently lack a written form.

2. Development of Foundational and Generative AI Models

- Build accurate and robust inclusive AI models for Speech Recognition (ASR), Speech Synthesis (TTS), Machine Translation (MT), and Optical Character Recognition (OCR) for all target tribal languages.
- Build advanced script conversion and transliteration tools to bridge dialectal variations and standardize digital communication.
- Develop advanced Generative AI models to be grounded in a verified, community-approved knowledge base, ensuring all outputs are factually accurate, culturally sensitive, and contextually appropriate.

3. Integration into National Public Services Platforms & Creation of Digital Tools

- Develop and deploy practical applications such as speech-to-speech translation, voice-based public service helpdesks, and AI assistants adapted for healthcare, education, and citizen services.
- Integrate Adivaani's technological stack with national platforms, including DIKSHA, Ayushman Bharat Digital Mission (ABDM), and PMGDISHA, to ensure broad adoption and last-mile impact.

2. Methodology: A Community-Integrated Development Pipeline

Adivaani uses an interdisciplinary, multi-layered approach that combines field linguistics, technology development, and strong community participation. This approach ensures that technological work is technically sound, ethically grounded, and socially accountable.

2.1. Community-Driven Data Curation

We will partner with Tribal Research Institutes (TRIs), universities, and trusted NGOs for co-creation and expert review at every stage of data collection, validation, and deployment. It will emphasize participatory governance.

Data collection will be prioritized across:

- **Conversational Speech:** Everyday dialogues, oral histories, and folk narratives to capture authentic language use.
- **Domain-specific discourse:** Terminology and ways of speaking used in agriculture, local governance, public health, and other essential sectors.
- **Cultural Expressions:** Songs, proverbs, and folklore that reflect the stylistic and semantic richness of each language.

2.2. AI Model Development and Ethical Alignment

Our technical pipeline is designed to develop foundational and generative AI tools for India's linguistic landscape:

- **Speech-to-Text (ASR):** We will leverage transfer learning from related high-resource Indian languages to adapt state-of-the-art multilingual ASR models. This will enable the accurate and large-scale transcription of invaluable oral traditions.
- **Text-to-Speech (TTS):** Utilizing expressive and narrative data, our TTS systems will employ advanced neural synthesis architectures to generate natural, human-like digital voices, powering accessible learning materials and communication aids.
- **Machine Translation (MT):** We will construct robust bidirectional translation systems between tribal languages and official languages (e.g., Hindi, English), fostering greater inclusion in governance, media, and public services.
- **Culturally-Grounded Generative AI:** A pioneering initiative of this mission is the development of Generative AI specifically architected for low-resource environments.
- **Language Model Alignment and Safety:** Models will be trained to respect Indigenous perspectives, to avoid bias, and to produce contextually appropriate outputs. Implementing structured feedback loops to monitor model responses, reduce misinformation, and ensure culturally respectful communication.
- **Human feedback loops:** We will use repeated guidance from community experts to improve model behaviour and ensure cultural sensitivity.
- **Multimodal Integration:** Systems will combine speech and text so they can translate, transcribe, and carry on simple conversational interactions.

2.3. Core Applications & Use Cases

The technologies and resources developed through Adivaani will be applied across critical sectors to strengthen socio-economic inclusion and improve daily life for tribal communities.

- **Health and Public Welfare:** In direct support of the **Ayushman Bharat Digital Mission (ABDM)**, our tools will eliminate language barriers between healthcare providers and patients. Real-time translation of medical instructions and public health advisories will make vital health information accessible in remote locations.

- **Education and Literacy:** Aligned with the **National Education Policy (NEP)** 2020’s emphasis on mother-tongue education, Adivaani will create a rich digital learning ecosystem. AI-powered tools integrated into the DIKSHA platform will help educators to create localized audiobooks, interactive lessons, and teaching aids, transforming the mother tongue into a bridge to modern learning.
- **Cultural Heritage Preservation:** We will create a secure National Tribal Language Digital Archive to preserve oral histories and cultural narratives for future generations. Our Generative AI tools will help communities transcribe, translate, and adapt traditional stories into new educational formats.
- **Agriculture and Livelihoods:** To empower farmers and artisans, we will develop multilingual advisory systems and voice-enabled chatbots. These tools will deliver real-time translations of bulletins from **Krishi Vigyan Kendras (KVKs)** and provide localized answers to queries on crop management, market access, and government schemes.
- **Governance and Digital Rights:** To ensure equitable access to citizen-centric services, we will build multilingual voice interfaces and helpdesk assistants for e-governance portals. These systems will allow tribal citizens to access welfare services, complete applications, and exercise their digital rights more effectively.

2.4. Ethical Governance and Data Sovereignty

Adivaani is built on a firm commitment to consent, transparency, accountability, and community benefit. The ethical framework rests on several core principles.

- **Free, Prior, and Informed Consent (FPIC):** All data collection will take place only after clear, free, prior, and informed consent from communities. Communities will retain co-ownership and access rights to their data.
- **Privacy and Security:** Sensitive data will be anonymized and stored under secure protocols, with explicit permission for public dissemination.
- **Community Ownership and Benefit Sharing:** Tribal communities will be recognised as the sovereign owners of their linguistic knowledge. Formal mechanisms will be established to ensure fair benefit sharing and to acknowledge the contributions of knowledge holders.

3. Implementation Strategy (5-Year Phased Execution)

The mission will be executed over five years through parallel, integrated tracks to maximize efficiency and accelerate impact:

- **Track 1: Data Curation & Community Engagement (Years 1–5):** Continuous scaling of data collection efforts to new languages and geographical regions.
- **Track 2: AI Research & Model Development (Years 1–5):** Iterative development, training, validation, and ethical alignment of all foundational and generative AI models.
- **Track 3: Application Deployment & Platform Integration (Years 1–5):** Collaborative deployment with partner ministries to embed and scale Adivaani tools within national digital infrastructure.

Financial Outlay (*Rs.30 Crore* over 5 Years)

Component	Allocation (<i>Rs. Crore</i>)	Strategic Purpose
A. Research & Development	15.0	Model Development & Generative AI research.
B. Data Collection & Field Ops	1.0	Support for field linguists and training workshops.
C. Technology Infrastructure	7.0	Secure, sovereign cloud servers, data storage, and maintenance of the Adivaani platform.
E. Platform & Application Development / Application Integration	6.0	Portal, Mobile Apps, piloting, API development, and integration of tools with DIKSHA, ABDM, and other platforms.
F. Monitoring & Contingency	1.0	Third-party evaluation, independent audits, ethical oversight, and unforeseen project needs.
Total (5-Year Investment)	<i>Rs.30.0 Crore</i>	A strategic investment in an equitable and inclusive digital future for India.

National Significance - Adi Vaani directly contributes to :

- **Digital India:** By extending the frontiers of digitization to our nation’s most marginalized communities, ensuring no one is left behind.
- **National Education Policy (NEP) 2020:** By providing the core technological infrastructure required to deliver the policy’s vision for mother-tongue-based multilingual education.
- **Atmanirbhar Bharat:** By developing sovereign AI capabilities tailored to India’s unique linguistic diversity, reducing dependence on foreign technologies and creating a model for the world.
- **Global Leadership:** Adivaani positioning India as a global leader in developing ethical, human-centric, and inclusive AI for linguistic preservation, in full alignment with UNESCO’s International Decade of Indigenous Languages (2022–2032).

Conclusion: From Ancestral Wisdom to a Digital Future

Language is the foundation of identity, history, and shared knowledge. For millions of India's tribal citizens, losing their mother tongue would mean losing vital collective memory and culture. Adivaani is therefore more than a technology project; it is a national responsibility and an opportunity for cultural renewal. By enabling tribal communities to speak, learn, and engage in their native languages in both physical and digital spaces, Adivaani will build a strong bridge between ancestral wisdom and future innovation. This mission is a decisive step toward an **inclusive, empowered, and truly Atmanirbhar India in this important Amrit Kaal.**